Name – Mohammad Saqib ali

Reg-NO – 12506923

Problem Number – 84

Problem Name – Largest Rectangle in Histogram

Problem level – Hard

```
class Solution {
public:
    int largestRectangleArea(vector<int>& heights) {
        stack<int> st;
        int maxArea = 0;
        int n = heights.size();

        for (int i = 0; i <= n; i++) {
            int h = (i == n) ? 0 : heights[i];

            while (!st.empty() && heights[st.top()] > h) {
                int height = heights[st.top()];
                st.pop();

                int width = st.empty() ? i : i - st.top() - 1;

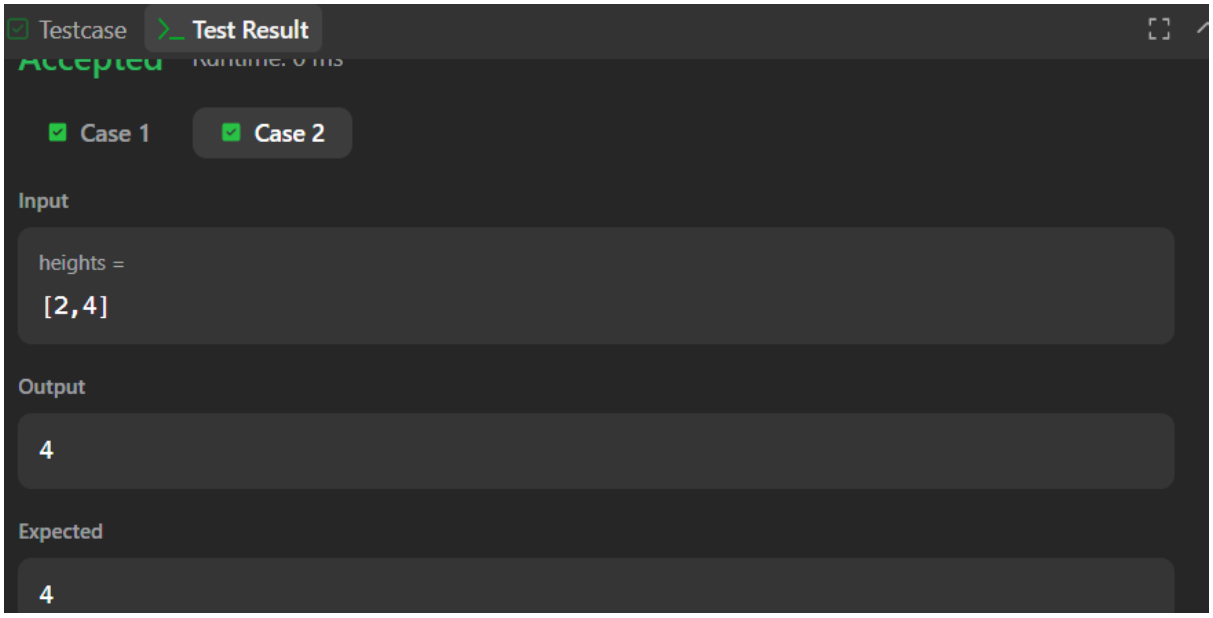
                maxArea = max(maxArea, height * width);
            }

            st.push(i);
        }

        return maxArea;
    }
```

};


Testcase
Test Result
Case 1
Case 2
Input
heights =
[2,1,5,6,2,3]
Output
10
Expected
10


Testcase
Test Result
Case 1
Case 2
Input
heights =
[2,4]
Output
4
Expected
4